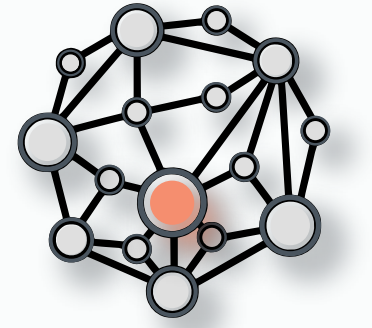
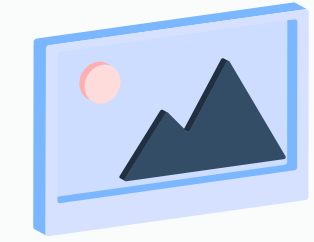
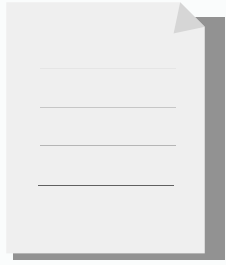




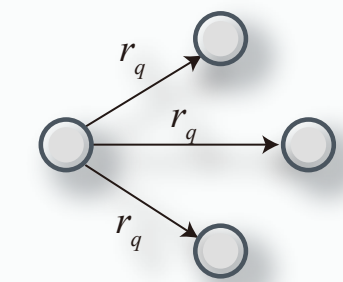
Knowledge graph



Visual modality



Textual modality



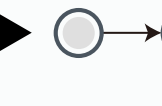
Query association



Rule mining



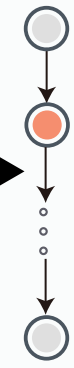
Horn rules



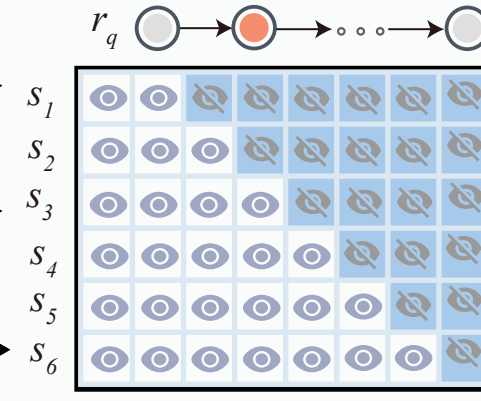
Triple pattern



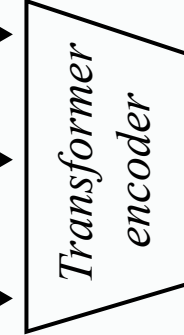
Pre-search



(A) Rule-Guided Reasoning Policy Network



Causal masked Transformer



Transformer
encoder



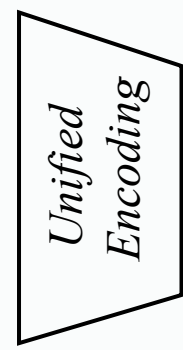
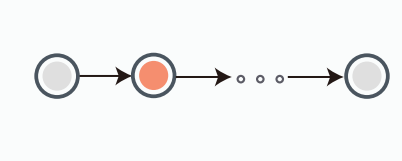
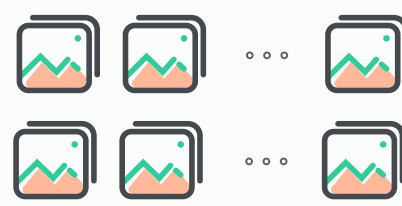
$\pi_\theta(a_t | s_t)$



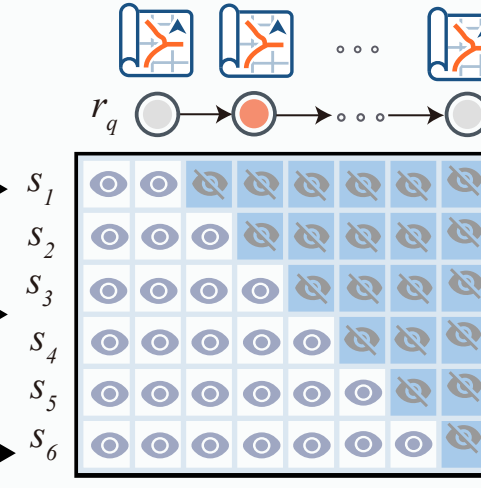
$\mathcal{L}_{RePN}$



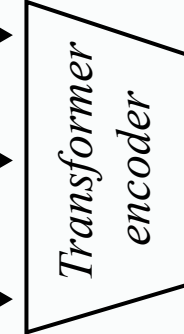
Tokenization



Unified
Encoding



Causal masked Transformer



Transformer
encoder



$\pi_\theta(a_t | s_t)$



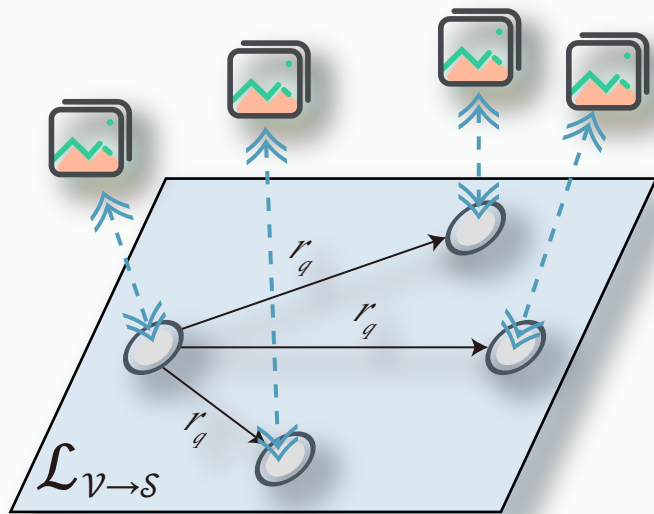
$\mathcal{L}_{MoIR}$



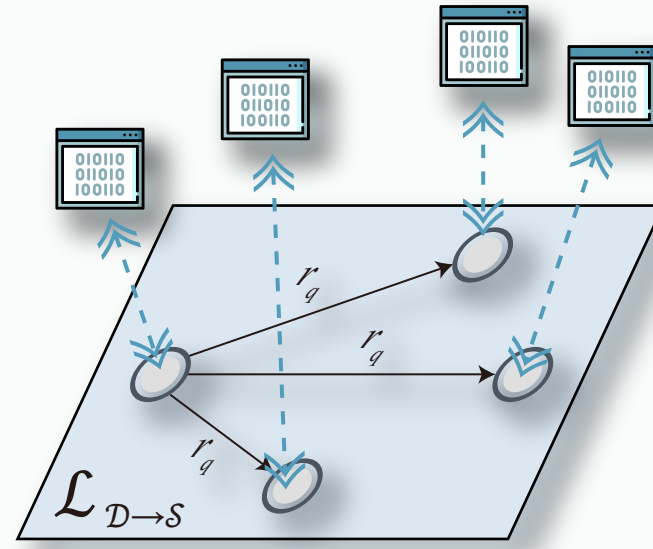
Tokenization



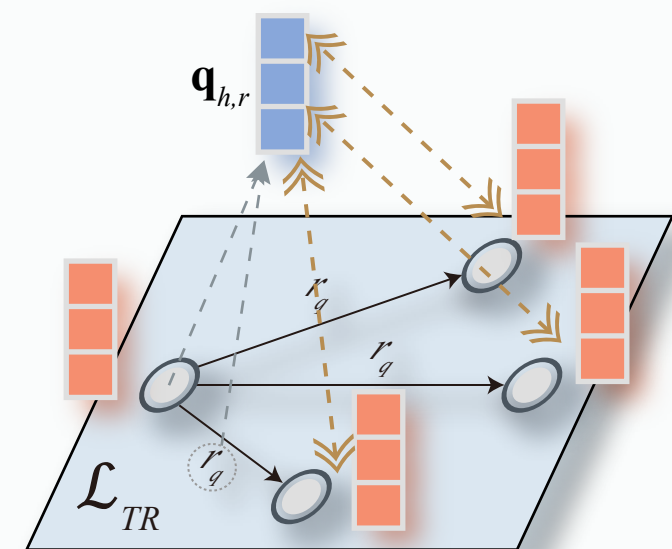
(B) Multi-modal Implicit Regularization Branch



$\mathcal{L}_{V \rightarrow S}$



$\mathcal{L}_{D \rightarrow S}$



$\mathcal{L}_{TR}$



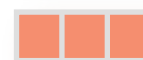
$\mathcal{L}_{CoSCL}$

(C) Cross-modal Semantic Contrastive Learning

$$r_q(X, Y) \leftarrow \bigwedge_{i=1}^L r_i(Z_{i-1}, Z_i)$$

 /  visual / textual tokens

 query representation

 entity representation

 /  contrastive learning

 rule-guided path

 query encoding

 modal fusion tokens